APPENDIX

A PROOF OF LEMMA 1

Define

$$\phi(x) = -\ln f_K(x). \tag{A1}$$

According to the log-concavity of $f_K(x)$, $\phi(x)$ is convex and $\phi'(x)$ is nondecreasing. For $z > \delta$, the likelihood ratio is

$$\Lambda(z) = \frac{f_K(z-\delta)}{f_K(z)}. \tag{A2}$$

Its logarithm can be expressed as

$$\ln \Lambda(z) = -\phi(z-\delta) + \phi(z). \tag{A3}$$

Differentiating with respect to z gives

$$\frac{\mathrm{d}}{\mathrm{d}z} \ln \Lambda(z) = \phi'(z) - \phi'(z-\delta) \geq 0. \tag{A4}$$

Therefore, $\Lambda(z)$ is nondecreasing for $z > \delta$. For $0 \leq z < \delta$, $\Lambda(z) = 0$, and hence $\Lambda(z)$ is nondecreasing over its entire support for $K > 0$.

When $K = 0$, the Rician power density reduces to the unit-mean exponential density,

$$f_0(x) = e^{-x}, \qquad x \geq 0. \tag{A5}$$

Accordingly,

$$\Lambda(z) = \frac{e^{-(z-\delta)}}{e^{-z}} = e^{\delta}, \qquad z \geq \delta, \tag{A6}$$

which is constant and therefore nondecreasing. Thus, for all $K \geq 0$, the likelihood ratio is nondecreasing in z, and the optimal likelihood-ratio test reduces to a single-threshold radiometer. ■

B PROOF OF THEOREM 1

For any fixed $P_J > 0$, the score of candidate jammer J_k is given by

$$Y_k(P_J) = \frac{P_J L_{J_k w}}{P_J L_{J_k b} X_{J_k b} + \sigma_b^2}. \tag{B7}$$

Since $\sigma_b^2 > 0$, $Y_k(P_J)$ is finite for every candidate jammer. Moreover, the candidate set $\mathcal{O}$ is nonempty and finite. Therefore, the finite score set

$$\{Y_k(P_J) : k \in \mathcal{O}\} \tag{B8}$$

always attains its maximum, which guarantees the existence of an optimal jammer.

If the joint distribution of the candidate scores is absolutely continuous with respect to the Lebesgue measure, then

$$\Pr\big(Y_i(P_J) = Y_j(P_J)\big) = 0, \qquad i \neq j. \tag{B9}$$

Since $\mathcal{O}$ is finite,

$$\begin{aligned}\Pr\big(\exists\, i,j \in \mathcal{O},\ i \neq j : Y_i(P_J) = Y_j(P_J)\big) \\ \leq \sum_{\substack{i,j\in\mathcal{O} \\ i<j}} \Pr\big(Y_i(P_J) = Y_j(P_J)\big) = 0.\end{aligned} \tag{B10}$$

Hence, ties among the candidate scores occur with probability zero, and the optimal jammer is almost surely unique. ■

C PROOF OF THEOREM 2

For each $P_J \in \mathcal{P}_J$, the preceding best-response analysis provides the corresponding optimal responses $P_a^*(P_J)$, $D^*(P_J)$, and $\lambda^*(P_J)$. If a response admits multiple optimizers, a fixed tie-breaking rule is used to define the response entering $G(P_J)$. Substituting these responses into the leader's utility reduces the optimization problem to

$$\max_{P_J \in \mathcal{P}_J} G(P_J). \tag{C11}$$

Since $\mathcal{P}_J$ is nonempty and compact and $G(P_J)$ is continuous on $\mathcal{P}_J$, the Weierstrass extreme value theorem guarantees the existence of at least one global maximizer. Thus, there exists $P_J^* \in \mathcal{P}_J$ such that

$$G(P_J^*) = \max_{P_J \in \mathcal{P}_J} G(P_J). \tag{C12}$$

The corresponding responses are optimal at their respective decision stages, and the leader cannot improve its utility by selecting another feasible jamming power while anticipating these responses. Therefore, P_J^* together with its corresponding optimal responses constitutes a Stackelberg equilibrium under the specified response rule.

We next establish the conditional uniqueness stated in Theorem 2. Suppose that $G(P_J)$ has a unique global maximizer and that the corresponding optimal responses are unique. Any equilibrium under the specified response rule must maximize $G(P_J)$ and therefore must use the same jamming power P_J^*. By the assumed uniqueness of the corresponding responses, it must also yield the same $P_a^*(P_J^*)$, $D^*(P_J^*)$, and $\lambda^*(P_J^*)$. Hence, the Stackelberg equilibrium is unique. When the uniqueness assumptions hold almost surely with respect to the channel randomness, this conclusion also holds almost surely.

Remark. If $\mathcal{P}_J$ is convex, strict concavity of $G(P_J)$ is a sufficient condition for the uniqueness of its global maximizer. The proof above does not require strict concavity; it uses the uniqueness assumptions stated in Theorem 2. Uniqueness of the corresponding optimal responses is additionally required for equilibrium uniqueness. ■